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RESEARCH ARTICLE

Edge-Federated Learning-Based Intelligent Intrusion Detection System for Heterogeneous Internet of Things

SHALAKA S. MAHADIK^{ID}, (Student Member, IEEE), PRANAV M. PAWAR^{ID}, (Member, IEEE), AND RAJA MUTHALAGU^{ID}

Birla Institute of Technology and Science, Pilani, Dubai Campus, Dubai, United Arab Emirates

Corresponding author: Pranav M. Pawar (pranav@dubai.bits-pilani.ac.in)

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ABSTRACT Distributed denial of service (DDoS) is an awful cyber threat, becoming more prevalent with mature heterogeneous IoT (HetIoT) applications like intelligent agriculture, wearables, and self-driving cars. Developing intelligent intrusion detection systems (IDS) using deep learning (DL) techniques to protect HetIoT has sparked a lot of attention. Federated learning (FL) helps to train the IDS locally with local data while respecting data privacy. Edge computing (EC) enhances security by processing data closer to the edge network. Therefore, the research contributed by integrating EC, FL, and DL and the proposed Edge-FL-based IDS. The proposed Edge-FL-based IDS aims to enhance HetIoT security by safeguarding data privacy against DDoS attacks. The research developed a DL-based convolutional neural network (CNN) classifier and used the CICDDoS2019 dataset to evaluate the success rate of the proposed Edge-FL-based IDS against DDoS attacks. The research employed IID and non-IID data distributions with participant clients $K = 3$, $K = 5$, and $K = 7$. The findings indicate that the proposed Edge-FL-based IDS outperforms centralized and other state-of-the-art FL models. The proposed Edge-FL-based IDS correctly detects and classifies DDoS attacks with the following accuracy: (a) 8-class IID: $K = 3$ is 99.98%, $K = 5$ is 99.97%, $K = 7$ is 99.96%. (b) 8-class non-IID: $K = 3$ is 99.97%, $K = 5$ is 99.94%, $K = 7$ is 99.90%. (c) 12-class IID: $K = 3$ is 99.96%, $K = 5$ is 99.96%, $K = 7$ is 99.95%. (d) 12-class non-IID: $K = 3$ is 99.07%, $K = 5$ is 96.44%, $K = 7$ is 52.39%.

INDEX TERMS Heterogeneous IoT, distributed denial of service (DDoS), federated learning (FL), independent and identically distributed (IID), non-IID, edge computing (EC).

I. INTRODUCTION

HetIoT technology includes heterogeneous sensors, protocols, and network architectures for the development of various applications, such as IoMT [1], IIoT [2], Green IoT [3], IoV [4], and many more. The emphasis of these applications is expanding significantly as the global community moves towards 6G technology, which will offer

faster connectivity and a high data transfer rate. However, the advent of technology gives birth to various cyber threats, rendering security and privacy an uphill battle [5], [6].

According to various researchers [7], [8], [9], the connectivity of HetIoT devices will surpass a value of \$700 billion by 2030. DDoS attacks are a significant cyber threat that has garnered considerable attention with the growth of HetIoT device connectivity. As reported by Help AG [10], the top DDoS attack vectors observed in 2021 were DNS, TCP, NTP, and UDP. Cloudflare [11] highlights that in the first quarter

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of 2023, cyber threats, including HTTP-DDoS and ransom-DDoS, targeted many nations. According to their analysis, the targeted organizations are gaming, cryptocurrency, telecommunications, and healthcare.

An IoT cybersecurity report [12] states that the number of infected IoT devices involved in botnet-driven DDoS attacks has increased from 200,000 to one million. Poor security measures in smart-connected devices, such as refrigerators, medical sensors, and smartwatches, motivate attackers to breach security and develop botnets [12]. It is hard to prevent the HetIoT devices from being part of a DDoS botnet, and tracing their origins is also challenging. Hence, implementing a defensive strategy is the most efficient way [13].

AI technologies have gained traction in developing intelligent IDS using ML and DL approaches. Over the millennia, intelligent IDS based on DL techniques have proven effective in detecting and classifying DDoS attacks and safeguarding HetIoT applications [9], [14], [15], [16].

The existing state-of-the-art work emphasizes designing IDS using centralized or distributed on-site learning approaches [17]. In the (a) *centralized learning* approach, a central entity (e.g., a cloud server) collects data from various HetIoT devices and trains the model with a centralized learning approach. The downside of this approach is that the data needs to travel through insecure communication channels, and the central entity has full access to all devices' raw data used to train the model. The raw data is network traffic containing vital information about the IoT devices. For example, the network traffic of wearable devices presents individuals with daily routines like physical movements. It raises concerns about data privacy and security. In (b) *distributed on-site learning* models, each HetIoT device will get the pre-trained models beforehand. Then, centralized learning happens on each device using their local data. Although this approach successfully handles security and privacy concerns, the first limitation is that HetIoT devices are resource-constrained and have limited power. In contrast, DL models are data-starving and need intensive computation to detect cyber threats like DDoS correctly. The second limitation is that to preserve data privacy, the devices refrain from exchanging data, impeding their ability to get updates regarding the most recent cyber threat trends.

The FL technique is a plausible solution to the downside mentioned above, which has recently piqued many researchers' interest. It outperforms traditional learning approach [1], [2], [3], [17]. It protects data privacy by 'bringing the model to the data, rather than the data to the model.' It scales efficiently, meets regulatory requirements such as GDPR and HIPPA, and supports real-time learning. Conversely, centralized learning is less adaptable for managing distributed environments and raises concerns about data privacy. The FL technique uses distributed data to perform decentralized learning, also term as *FL-based decentralized learning* [18]. FL's cyclic incremental learning facilitates continuous learning, keeping each HetIoT device updated with recent cyber threats. Such learning is known as *FL-based*

distributed on-site learning. Fig. 1 compares the traditional learning approach and the FL approach using both centralized and distributed on-site learning. It is pertinent to note that *FL-based distributed on-site learning* is beyond the scope of this research. The remaining research will focus on the *FL-based centralized learning*.

EC served as a better alternative to cloud computing in recent years [5], [19], [20], [21]. Cloud servers in geographically remote places encounter latency and bandwidth consumption issues. An application like a self-driving car demands a fast response with minimal latency. In an emergency, the remote monitoring system demands instant data to diagnose a patient's disease correctly. EC allows data processing to take place near the edge network. It reduces latency time and bandwidth requirements while strengthening security [22], [23], [24]. Therefore, adopting EC is a viable solution to boost further the security of the FL-based intelligent IDS that brings the model and processing server near the network [9], [14].

A. MOTIVATION AND OBJECTIVE

The research is motivated to delve deeper into the DDoS attack and provide countermeasures to safeguard the HetIoT applications. The research adopted EC to deploy the proposed IDS near the edge network and train the model on a distributed environment using FL. The research leads to the development of a DL-based one-dimensional CNN classifier, i.e., a DL-based 1D CNN classifier. Thus, the research leverages EC, FL, and DL to improve HetIoT security and privacy. The main objective of the research is to *develop an Edge-FL based intelligent IDS to protect HetIoT applications like IoMT, IIoT, IoV, Agri 4.0-IoT, and many others from DDoS cyberthreat*.

B. PROBLEM DEFINITION

There are several issues that arise when FL trains the model using local data from HetIoT devices. A lot of heterogeneous data, is produced by HetIoT applications. It is one of the main issues with the FL approach, as it leads to IID and non-IID data distribution concerns [25]. The FL approach performs well with IID data distribution, where each client holds equal traffic data. However, there is a considerable decline in performance when dealing with non-IID data distribution [26], [27]. Each client holds unbalanced traffic data in non-IID distribution. The research focused on such problems while developing an IDS. The research aims to create an IDS to effectively manage data heterogeneity by addressing IID and non-IID data distribution issues.

C. CONTRIBUTION

The key contribution of the research is listed below.

- The research combined FL with EC, addressing security and privacy concerns. It will reduce the data processing load on the central entity. Moreover, the research provides insights into the recent state-of-the-art FL models.

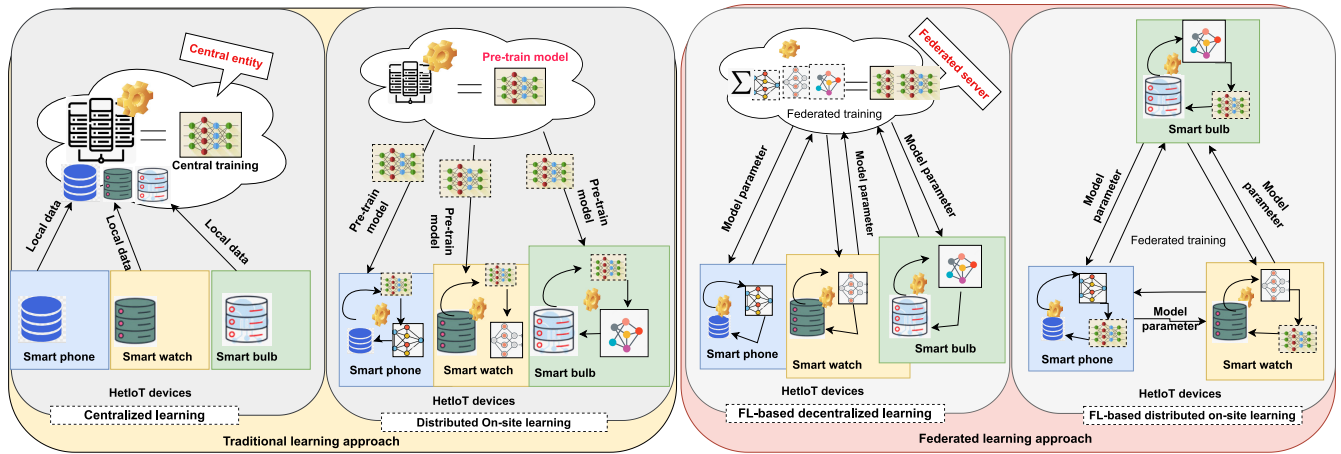


FIGURE 1. General working of Traditional learning and Federated learning.

- The research employed the CICDDoS2019, a real-time benchmarking cybersecurity dataset, to develop a proposed Edge-FL-based IDS to protect HetIoT applications against DDoS threats. The research details the data preprocessing steps and the creation of two datasets, Dataset-8 and Dataset-12, to perform 8 and 12 class classifications, respectively.
- The research includes the creation of a DL-based 1D-CNN classifier used as a base model for developing the proposed Edge-FL-based IDS. The classifier is lightweight in terms of the convolutional layer and filters employed compared to existing state-of-the-art IDS.
- The research examine the performance of the proposed Edge-FL-based IDS and Centralized IDS against Dataset-8 and Dataset-12. Furthermore, the research analyze the performance of the proposed Edge-FL-based IDS under IID and non-IID data distributions, with clients $K = 3$, $K = 5$, and $K = 7$.
- The research shows that compared to Centralized and state-of-the-art FL models, the proposed Edge-FL-based IDS performs well while maintaining data privacy and honing security in the HetIoT context. The proposed Edge-FL-based IDS detects and classifies DDoS cyber threats with the following accuracy:
 - (a) 8-class IID: $K = 3$ is 99.98%, $K = 5$ is 99.97%, $K = 7$ is 99.96%.
 - (b) 8-class non-IID: $K = 3$ is 99.97%, $K = 5$ is 99.94%, $K = 7$ is 99.90%.
 - (c) 12-class IID: $K = 3$ is 99.96%, $K = 5$ is 99.96%, $K = 7$ is 99.95%.
 - (d) 12-class non-IID: $K = 3$ is 99.07%, $K = 5$ is 96.44%, $K = 7$ is 52.39%.

D. ROADMAP

The remainder of this article is organized as follows: Section II includes existing state-of-the-art FL models. Section III includes the research methodology by outlining

dataset details, a DL-based 1D-CNN classifier, the proposed Edge-FL-based IDS framework, and implementation details. Section IV includes results and discusses IID and non-IID data distribution with 8 and 12-class under Centralized and the proposed Edge-FL-based IDS. Section V concludes the research with an outlook for potential future directions. Table I presents a list of abbreviations used in the article.

II. RELATED WORK

This section focuses mainly on recent work on developing AI-based IDS that employs an FL-based approach to protecting networks from anomalies.

A hybrid ensemble-based IDS that aims to enhance FPR and TPR to mitigate the occurrence of network outages has been proposed by the author [25]. The proposed method performs well in TPR and FPR in both noisy and clean data environments. The proposed method included four IoT network threats, such as DoS, Probe, R2L, and U2R, and obtained individual class accuracy rates as 92.05%, 99.98%, 98.38%, and 98.71% in centralized environments and 88.42%, 98.27%, 98.71%, and 93.74% in FL environments.

The development of IDS using neural networks has gained popularity. However, models developed using such classifiers are more susceptible to malicious attacks. So, author [33] presented an IDS based on a GBDT algorithm using data binning and masking techniques. Instead of the FedAvg aggregation function, the proposed method masks the feature data, and the unique count of feature data bins is shared with the server. The performance evaluation used a centralized model with FedAvg algorithm training, and the proposed method used a data bin. When $K=21$, the proposed method achieves 65.04% accuracy, and for $K=66$, it achieves a 64.52% accuracy rate.

Each IoT device has its security mechanism, but once connected to a network, a network vulnerability to one IoT device may signify a threat to many. To address this issue, the author [28] proposed FL using an ML algorithm to secure

TABLE 1. List of Abbreviations.

Acronym	Definition
AE	Autoencoder
AI	Artificial intelligence
CCPA	California consumer privacy act
DNS	Domain name system
EDA	Extensive data analysis
FN	False negative
FP	False positive
FPR	False positive rate
GAN	Generative adversarial networks
GBDT	Gradient-boosting decision tree
GDPR	General data protection regulation
GRU	Gated recurrent unit
IIoT	Industrial IoT
IoMT	Internet of medical things
IoT	Internet of things
IoV	Internet of vehicles
LDAP	Lightweight directory access protocol
LR	Logistic regression
LSTM	Long short-term memory
ML	Machine learning
MLC	Multi-label classification
MLP	Multilayer perceptron
NetBIOS	Network basic input-output system
Non IID	Not independent and identically distributed
NTP	Network time protocol
PCC	Pearson correlation coefficient
PII	Personally identifiable information
R2L	Remote-to-local
SDN	Software-defined network
SNMP	Simple network management protocol
SSDP	Simple service delivery protocol
TCP	Transmission control protocol
TFTP	Trivial transfer file protocol
TN	True negative
TP	True positive
TPR	True positive rate
U2R	User-to-root
UDP	User datagram protocol

IoT devices from network cyber threats using the NSL-KDD dataset. The proposed approach considers three cases: the first and second involve four client devices training with non-IID and IID data distribution, and the third involves eight client devices with IID data distribution. LR performs better than MLC and achieves a global accuracy of 85% during the second case. The average global accuracy reported by the author is 94% on unseen tested datasets.

Federated IDS using the GRU technique for various DDoS attack detection has been proposed by author [38]. The proposed model performance has been assessed in two scenarios, i.e., using FedAvg as an aggregation function and using prototypical weight extraction. Local training is 2000 rounds, whereas global server training is 10 rounds. Syn, MSSQL, UDP-Lag, LDAP, network basic input/output system (NetBIOS), UDP, and benign classes from CICDDoS2019 datasets are considered to evaluate the proposed method.

Another author [35] proposed a residual neural network based on one-dimensional CNN for network traffic analysis. The proposed approach efficiently prevents gradient dispersion while training the network. The study considers

six DDoS classes, namely, MSSQL, NTP, TFTP, UDP, LDAP, NetBIOS, and Benign traffic from the CICDDoS2019 dataset, to evaluate the model. The proposed model attained 99.20% accuracy with 600 communication rounds.

An FL-based solution using the MLP technique is offered by author [32] to protect the software-defined network against DDoS attacks. The proposed method considers the CICDDoS2019 dataset with six classes, namely NetBIOS, LDAP, MSSQL, UDP, TFTP, and NTP, and achieves 98.84% accuracy after 30 communication rounds with four participant clients. The proposed method divides the dataset across six clients, each of which holds all six classes specified above in a balanced manner.

Centralized IDS built with DL methods have limited data storage capacity, incur high computing costs, and pose data privacy concerns. To overcome this difficulty, author [19] created a self-attention network and CNN classifier hybrid in an FL environment using the NSL-KDD dataset. DoS, Probe, R2L, and U2R are the IoT network threats investigated, and the proposed hybrid classifier's testing accuracy reported is 92.21% after 50 training rounds.

Next, an anomaly detection model using a deep AE approach has been proposed by author [23]. The proposed model considers deployment near network security gateways, aiming to protect data privacy and security. The author considers the IoT botnet dataset, which contains TCP, UDP, Syn, and ACK flooding attack classes. The proposed model's performance is assessed using binary classification with 50 federated rounds. The proposed model achieves 94.04% average accuracy using the FL approach, whereas 93.73% using the centralized approach.

Further, to detect DDoS cyber threats, author [31] proposed IDS using the MLP approach and the CICDDoS2019 dataset. The author considers 13 clients, each with an exact number of benign and DDoS class traffic data distributions. Thirteen DDoS attack class addresses are DNS, LDAP, MSSQL, NTP, SNMP, SSDP, NetBIOS, UDP, Syn, UDPLag, TFTP, Portmap, WebDDoS, and benign. The proposed model does better than the FedAvg algorithm regarding convergence time. It gets an F1-score of 98.53% after 605 global communication rounds and achieves a TPR of 99.77%.

Many IoT devices are available in the market with no security mechanism, and many more will be available as B5G technology develops. These devices are vulnerable within any network, posing a cyber threat. As a result, [29] proposed an IoT malware detection solution based on the DL-FL technique for the B5G network. The author considers four different classifiers with zero, one, two, and three hidden layers, respectively, for supervised learning.

The heterogeneous nature of IoT devices and the necessity for constant training for optimal performance make centralized ML-based IDS solutions challenging to implement. So, the author [30] presented a decentralized FL solution based on LSTM and GRU algorithms to overcome this concern. The proposed approach uses a random-forest-decision tree classifier as an ensemble to achieve a high accuracy rate.

The attacks addressed are Man-in-the-Middle attacks, Ping DDoS Flood attacks, Modbus Query Flood attacks, and Syn DDoS attacks. The overall accuracy mentioned in the study is 90.25% using the FL approach. Although FL is a promising solution to maintain data privacy, its local training needs diverse and more data to obtain an efficient model. The GAN is one way to augment and balance the training data. Therefore, [24] proposed distributed IDS using the GAN approach. The proposed approach considers 200 local training rounds and 20 global communication rounds.

DDoS attacks using botnets are becoming highly effective, with a considerable profit margin. IoT devices are more susceptible to cyberattacks. As a result, many intruders continue to target weak IoT devices. To overcome this barrier, [22] proposed an iterative model averaging with a GRU classifier. The proposed approach is 3.7 times costlier against DDoS cybercriminals. [34] proposed approach shows collaborative learning performs better than traditional centralized learning using a deep AE approach. The author divides the NSL-KDD dataset into a train set and a test set; the train set is further split into a train and validation set used during local model training. The proposed approach considers 20 communication rounds to achieve 98.06% accuracy for binary classification.

To protect the SDN environment, author [37] proposed FL-based DDoS threat detection using a hybrid of CNN and MLP approaches. The author considers two feature selection techniques, i.e., the PCC (24 features) and the Chi-Square technique (25 features). The CICDDoS2019 dataset was used to test the proposed model's performance. It looks at four classes: benign, DNS, UDP, and Syn. The proposed hybrid model achieves 98.18% accuracy with 25 selected features and 98.37% accuracy with 24 selected features at 30 global communication rounds.

The literature emphasizes solely the centralized FL approaches that maintain data privacy and boost the security of various HetIoT applications. Table 2 evidences the most recent state-of-the-art work to develop various IDS employing AI approaches, primarily focusing on FL, from 2020 to 2023. It highlights the various AI techniques and security datasets used. Additionally, it gives insights into the data distribution, i.e., IID or non-IID. Table 2 further shows the number of participant clients, aggregation strategies utilized, and FL libraries used during the FL training process. The model validation approach, feature extraction, and classification type, i.e., multiclass or binary, are presented. The purpose of the threat model and its broad application domain are also included in Table 2.

The existing state-of-the-art [31], [32], [33], [35], [37], [38] followed a similar approach as the research aims. Although the existing state-of-the-art work considers a centralized FL approach, some take data distribution as IID or non-IID with more communication rounds. For example, [31] takes 605 communications rounds, whereas [35] takes 600 communications rounds. Furthermore, except [33] and [31], all others consider only three or five classifications

for detecting and classifying the DDoS cyberthreat. Many existing state-of-the-art IDS provided good accuracy but failed to demonstrate the individual class performance required to ensure IDS performance. With more participant clients, the [33] achieved an accuracy rate of 65.04%, 64.52%. The [35] attained 99.20% accuracy with 600 communication rounds. Additionally, [37] considers the CNN-MLP technique with two stack convolutional layers and two parallel convolutional layers, each with eight filters. Two convolutional layers follow one max pooling layer. Another state-of-the-art IDS [35] considers a 4-residual convolutional layer with 32, 16, 8, and 4 filters. It indicates that present state-of-the-art models are complex in terms of the number of layers and neurons employed.

While addressing the above shortcomings, the research aims to contribute by designing intelligent IDS using the CICDDoS2019 dataset that contains the most recent DDoS attack classes. The research analyzes the performance of a proposed Edge-FL-based IDS with both IID and non-IID data distribution by considering 8-class and 12-class classifications. The research uses the validation technique to validate the proposed Edge-FL-based IDS performance, utilizing a validation round. The validation confirms that the proposed Edge-FL-based IDS is not biased and is functioning appropriately. Likewise, compared to [35] and [37], the research develops a simple and lightweight DL-based 1D-CNN classifier using three convolutional layers, each with 16, 16, and 32 filter sizes.

The research also uses EC to deploy intelligent IDS near the edge network, where the existing state-of-the-art lacks concentration. Positioning the proposed Edge-FL-based IDS near the edge network will assist the network administrator in promptly detecting and separating the DDoS traffic from the legitimate users within the network. Furthermore, the research featured individual class performance that helps network administrators discover the root cause of protocols such as UDP, SNMP, LDAP, Portmap, and SSDP. The admin blocks such traffic without disrupting the entire network, letting legitimate users access the services. The following section covers in detail the research methodology utilized throughout the research.

III. RESEARCH METHODOLOGY

This section outlines the research methodology in detail, including data preprocessing steps, the development of DL-based 1D-CNN classifiers, and the framework of the proposed Edge-FL-based IDS using dataset-8 and dataset-12 to detect and classify DDoS cyber threats in the HetIoT environment.

A. CICDDOS2019 DATASET

The CICDDoS2019 [39] security dataset is employed in the research, a publicly available benchmarking dataset. The Canadian Institute for Cybersecurity (CIC) and the University of New Brunswick (UNB) contributed to the dataset. There are many other security datasets available, including

TABLE 2. State-of-the-art IDS using FL approach (2020-2023).

Year	Threat model & Ref.	AI technique	Security Dataset	Data distribution approach	Participant clients	Aggregation technique	FL Library	Model Validation	Classification	Feature extraction	Application domain
2020	Intrusion detection [25]	KNN, RF and MLP	NSL-KDD, DSOS Traffic Traces, Gas Pipeline Dataset, Water Tank Dataset, CICDDoS2019	Unbalanced non-IID	K=4	FedStacking	—	No	4-Class	PCA	IoT applications
2021	DDoS [33]	GBDT		Balanced non-IID	K=21,66	Data masking and Binning technique FedAvg	—	No	14-Class	No	IoT applications
2021	Network threat detection [28]	LR, MLC	NSL-KDD	Unbalanced IID and non-IID	K=4,8	FedAvg	PySyft	No	5-Class	No	IoT applications
2021	DDoS [38]	GRU	CICDDoS2019	non-IID	K=5	FedAvg, Prototypical weight extraction	PyTorch	No	6-Class	No	IoT applications
2021	DDoS [35]	CNN	CICDDoS2019	IID, non-IID	K=6	FedAvg	—	Yes	6-Class	No	IoT applications
2022	DDoS [32]	MLP	CICDDoS2019	Balanced IID	K=3,4,5	FedAvg	—	No	6-Class	No	IoT applications IIoT, SDN
2022	Intrusion detection [19]	CNN	NSL-KDD	Unbalanced	—	FedAvg	TFF	No	4-Class	No	IoT applications
2022	Anomaly detection [23]	Deep AE	IoT botnet	Balanced	—	FedAvg	PySyft	No	2-Class	Yes	IoT, EC
2022	DDoS [31]	MLP	CICDDoS2019	Unbalanced non-IID	K=13	FLAD	TFF	Yes	13-Class	Yes	IoT applications
2022	Malware detection [29]	MLP, AE	N-Balot	—	K=8	Mini batch aggregation and Multi epoch aggregation	—	No	2-Class	No	IoT applications
2022	Anomaly detection [30]	GRU	Modbus	—	—	FedAvg	PySyft	No	Multiclass	No	IoT applications
2022	Intrusion detection [24]	GAN	NSL-KDD, KDD99, and UNSW-NB15	non-IID	—	FedAvg	—	No	2-Class, Multiclass	No	IoT, EC
2022	DDoS [22]	GRU	UNSW NB15	—	K=4	FedAvg	PySyft	No	Multiclass	No	IIoT, EC
2022	DDoS [37]	CNN-MLP	CICDDoS2019	IID	K=6	FedAvg	—	No	4-Class	PCC and Chi-square technique	SDN, IIoT
2023	Anomaly detection [34]	Deep AE	NSL-KDD	—	—	FedAvg	—	Yes	2-Class	No	IoT applications
2023	Cyber threat intelligence [36]	CNN	ISCX-IDS-2012, CICDDoS2019	—	—	FedAvg	—	No	4-Class	No	Blockchain
Proposed Edge-FL based IDS for DDoS		CNN	CICDDoS2019	Unbalanced IID and non-IID	K=3,5,7	FedAvg	TFF	Yes	8-Class, 12-Class	Random forest regressor	HetIoT, EC

NSL-KDD, TON_IoT, N-BaIoT, and others; however, the CICDDoS2019 dataset stands out as the sole dataset encompassing the highest number of distinct DDoS attack classes. The most common techniques of targeting HetIoT networks involve several DDoS attacks readily available in the CICDDoS2019 database (refer to Fig. 2). It is the primary rationale for selecting this dataset.

CICDDoS2019 dataset considers attacks similar to DDoS attacks that happen in IoT networks. For example, the Mirai attack is a large-scale DDoS that can target IoT devices. The purpose of a Mirai attack is to infect various computers, laptops, routers, and workstations connected to the internet. Such infected devices then continuously monitor the web for vulnerable IoT devices and turn them into an IoT botnet network to execute DDoS threats. For example, the Mirai botnet infected Dyn, a DNS service provider, in October 2016 [40], [41], [42]. Another example is the shutdown of intelligent heating systems in Finland in November 2016 owing to the Mirai botnet. Many such incidents exist, indicating that DDoS attacks can be launched using any vulnerable HetIoT devices to form an IoT botnet [43]. It is imperative to acknowledge that while the CICDDoS2019 dataset does not originate from an IoT environment, it comprises traffic from highly influential protocols standard to HetIoT contexts such as TFTP, DNS, and SSDP. Mirai malware frequently targets These protocols in HetIoT infrastructure. Subsequently, the dataset favors researchers whose primary emphasis is detecting DDoS cyber threats within the context of the HetIoT [16], [36]. Additionally, the dataset is widely recognized within the academic community for its utility in advancing different IDS [31], [32], [33]. Therefore, the research employed this dataset to detect and classify various DDoS cyber threats.

The author [39] described the network architecture and features in depth. The following is a summary of the CICDDoS2019 dataset: The dataset test-bed was divided into the victim-network and attack-network parts. The victim network comprises firewalls, routers, Ubuntu web servers, and Windows machines. The attack network is an independent network that executes various DDoS attacks via TCP and UDP-based protocols. Syn (*a.k.a. Protocol*

exploitation-based attack), MSSQL, and SSDP are examples of TCP-based attacks, (*a.k.a. reflection and Amplification-based attack*). In contrast, NTP and TFTP are examples of UDP-based attacks (*a.k.a. Reflection and Amplification-based attack*). The detailed classification taxonomy is depicted in Fig. 2.

The dataset is entirely labeled as benign and various DDoS attack classes, and it contains 80 network-related features. The dataset is available in both PCAP and CSV file formats. The dataset is in two separate folders: the training folder with *Eleven-CSV files* and the testing folder with *Seven-CSV files*. Each CSV file includes one or two DDoS attacks and benign traffic with over three million instances. The *TFTP.CSV* file includes 20,107,827 instances with TFTP attack class and benign traffic. Some CSV files, e.g., *Portmap.CSV*, include only 191,694 instances with Portmap attack traffic and benign traffic, whereas *UDP-lag.CSV* consists of 725,165 instances with UDP-Lag, Syn, and UDP attack classes and benign traffic.

DATA PREPROCESSING

This segment explores the data preprocessing steps used to create two distinct datasets, namely, *Dataset-8* and *Dataset-12*. For any AI model to perform better, EDA is required. It assists the model in reducing training time by ensuring that it does not overfit or underfit, hence expediting good performance. The network security datasets contain an enormous number of features. The AI classifier's performance degrades as the number of features rises. So, the research used the data preprocessing steps represented in Fig. 5 and elaborated below.

- The dataset is present in two separate folders with *Eleven-CSV files* and with *Seven-CSV files*. The *Eleven-CSV* files contain 48,930,145 samples, including eleven different types of DDoS attacks and benign traffic. The *Seven-CSV* files contain 20,364,525 samples of DDoS attack traffic and benign traffic.
- As a part of data cleaning, each CSV file is read one by one, and notice that each CSV file contains infinite (inf, -inf), null, and duplicate values that are removed.
- Each CSV file contains features like Flow Id, Source-IP, Destination-IP, SimillarHTTP, Timestamp, Source-port, Destination-port, and Unnamed, which are non-informative are removed [16], [44].
- An effective feature selection technique is essential to reduce computational costs and elevate IDS performance [37]. So, the research applied the *RandomforestRegressor* [39], [45] algorithm on each CSV file to pick out the most essential features. Following that, the features are sorted from most important to least important.

Fig. 3 and Fig. 4 show the most relevant features extracted from each CSV file and stored separately. Lastly, these CSV files with extracted important features are combined, and two separate datasets are created:

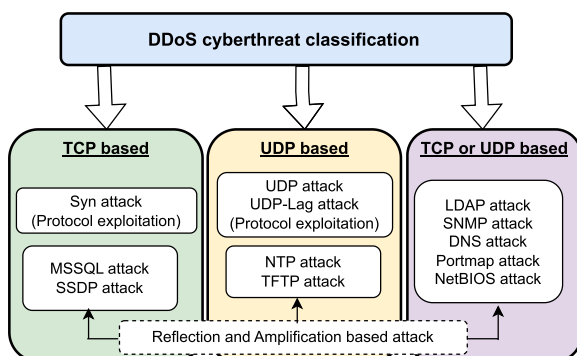


FIGURE 2. DDoS attack classification [5], [39].

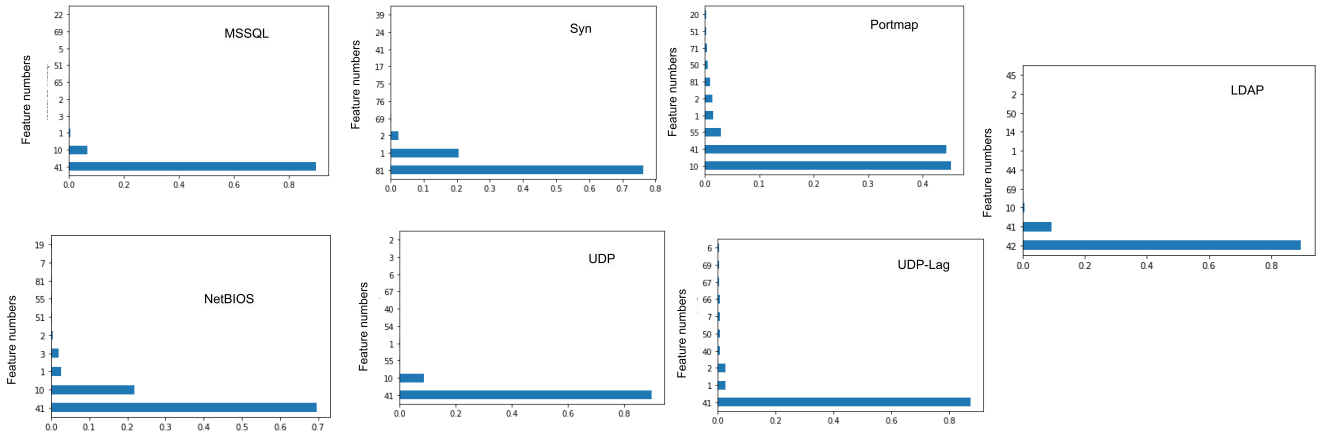


FIGURE 3. Important features for Seven-CSV files.

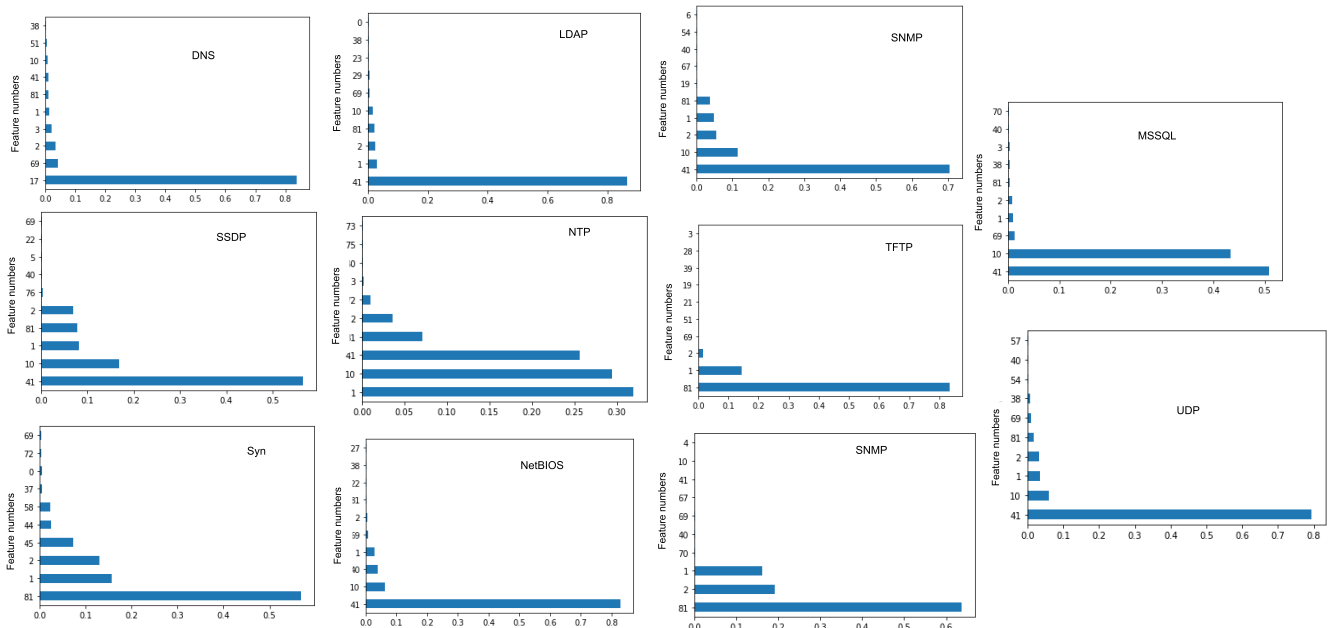


FIGURE 4. Important features for Eleven-CSV files.

Dataset-8 with 20 features and *Dataset-12* with 19 features. Table 3 lists the best-selected features for both datasets. The feature selection pseudocode is presented in Algorithm 1.

- The *Dataset-8* and *Dataset-12* are further examined to find if they contain any duplicate and NaN values after merging different CSV files; it is found that ‘NaN’ values are present in both datasets. So, all ‘NaN’ values are replaced by simple-imputer’s ‘constant’ strategy method [46], [47].
- Lastly, the research adopted these newly processed datasets, i.e., *Dataset-12* and *Dataset-8*. The details of these datasets, along with the class distribution, are provided in Table 4.

- The research included the ‘standard scaler’ normalization method to ensure that all features contribute equally to the classifier and that features with higher values do not take over. This strategy improves the learning process and helps the DL-based 1D-CNN classifier perform better by avoiding biased performance [44], [48].
- Both datasets are split into three categories: train set (60%), valid set (20%), and test set (20%). The validation set assists in fine-tuning IDS performance by increasing accuracy after each epoch [49]. Simultaneously, it ensures that the DL-based 1D-CNN classifier is not overfitting or underfitting and that training is going appropriately. *Dataset-12* contains all

Algorithm 1 Feature Selection.

Number of CSV Files in Each Folder Either $n = 11$ or 7

- 1 Read each CSV file as $file_1, file_2, \dots, file_n$ with features X
- 2 $RF_{model} \leftarrow$ apply standard-model with default settings(RF)
- 3 **while** $features(file_n) > X$ **do**
- 4 $train_file \leftarrow (file_n)$
- 5 Train RF_{model} with $train_file$
- 6 Sort with most relevant features (RF_{model})
- 7 Save into new file, i.e. ($file_n^{new}$)
- 8 **end**
- 9 Combine all the top selected features in the file, i.e., $file_1^{new}, file_2^{new}, \dots, file_n^{new}$

TABLE 3. List of best-selected features after Random Forest Regressor algorithm.

dataset-12		dataset-8	
Feature No	Best selected features	Feature No	Best selected features
1	Source Port	1	Source Port
2	Destination Port	2	Destination Port
3	Protocol	3	Protocol
10	Fwd Packet Length Min	6	Total Backward Packets
17	Flow Bytes/s	7	Total Length of Fwd Packets
29	Bwd IAT Mean	10	Fwd Packet Length Min
37	Fwd Header Length	20	Flow IAT Std
38	Bwd Header Length	40	Bwd Packets/s
40	Bwd Packets/s	41	Min Packet Length
41	Min Packet Length	42	Max Packet Length
44	Packet Length Std	44	Packet Length Std
45	Packet Length Variance	50	ACK Flag Count
51	URG Flag Count	51	URG Flag Count
58	Fwd Header Length.1	55	Average Packet Size
69	Init_Win_bytes_forward	66	Subflow Fwd Bytes
70	Init_Win_bytes_backward	67	Subflow Bwd Packets
72	min_seg_size_forward	69	Init_Win_bytes_forward
76	Active Min	71	act_data_pkt_fwd
81	Inbound	76	Active Min
		81	Inbound

TABLE 4. Processed dataset details.

Dataset-8	Label no.	Class distribution	Dataset-12	Label no.	Class distribution
Benign	0	51,888	Benign	0	33,912
LDAP	1	121,198	DNS	1	532,414
NetBIOS	2	1,095,333	LDAP	2	513,239
MSSQL	3	1,112,013	MSSQL	3	536,228
Portmap	4	177,197	NTP	4	538,034
Syn	5	973,289	SNMP	5	532,848
UDP	6	1,103,375	SSDP	6	539,390
UDPLag	7	1,873	NetBIOS	7	534,990
Total samples =		4,636,166	UDP	8	532,125
			Syn	9	552,149
			UDPLag	10	330,079
			TFTP	11	534,884
			Total samples =		5,710,292

DDoS classes in a balanced manner, so the research utilized a random sampling approach while splitting the dataset [50]. *Dataset-8* is highly unbalanced, so the research utilized the *stratify sampling* approach while splitting the dataset, which helps to maintain the exact

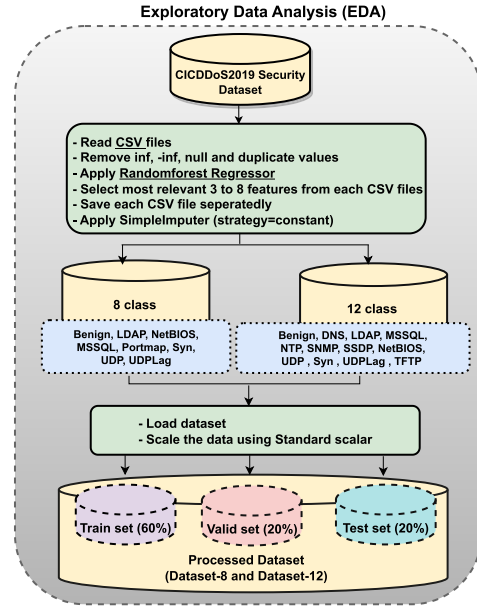


FIGURE 5. Data preprocessing steps.

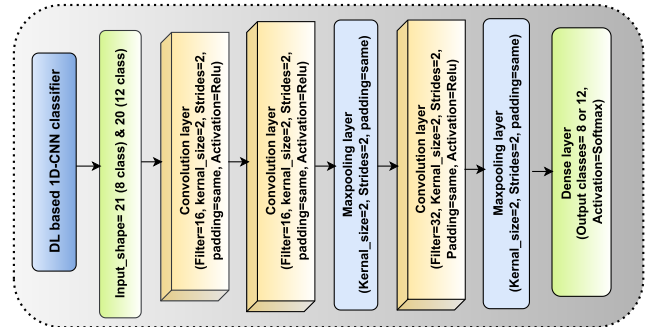


FIGURE 6. Architecture of DL-based 1D-CNN classifier.

proportions of each class across the splits. As a result, the training, validation, and test sets retain the original class distribution by including an appropriate portion of each class. It assists the DL-based 1D-CNN classifier in learning to identify trends and predict results across all classes, resulting in a more robust IDS [50].

After the train-valid-test split, both datasets are used in the research to develop centralized IDS and proposed Edge-FL-based IDS. The following section gives insight into the architecture of the DL-based 1D-CNN classifier.

B. DL-BASED 1D-CNN CLASSIFIER

DL-based CNN models are popular due to their potency in recognizing and analyzing patterns within images, computer vision, anomaly detection, and other applications. Likewise, these strategies are helpful in the detection and classification of network traffic analysis [18]. As a result, the research leveraged a CNN model to create a customized DL-based

1D-CNN classifier to detect and classify various DDoS threats in a HetIoT context.

The customized DL-based 1D-CNN classifier's architecture consists of three 1D-convolutional layers with 16, 16, and 32 filters for each convolutional layer, respectively, while the kernel size is 2 for all convolutional layers. Two max-pooling layers with kernel sizes and strides of two are adopted. A 'Softmax' activation function uses a single dense layer as the final output layer to detect and classify various DDoS attack classes and benign traffic. The research uses the 'Relu' activation function since it is substantially faster than other functions. The loss function considered is 'SparseCategoricalCrossentropy' as the target column is an integer, not one-hot encoded [16], [51]. The 'ADAM' optimizer is more efficient and potent than other optimizers, such as SGD and RMSprop. Its rapid adaptive learning and momentum-based strategy allow the classifier to train faster and converge more quickly to the optimal solution; consequently, it was adopted in the research.

The research employed *Grid search* technique in conjunction with thorough experimental analysis to choose the number of convolutional layers, filters, kernel size, and activation function. The research included various filters like 8, 16, 32, and 64; kernel sizes tried are 2,3,4; optimizers like SGD, RMSprop, and ADAM with a learning rate of 0.01 and 0.001 during the *trial-and-error* approach and *Grid search* technique. Further, the research also inputs activation functions such as Relu and Sigmoid and finds that 'Relu' is substantially faster and converges faster to the ideal solution than Sigmoid. The customized DL-based 1D-CNN architecture is shown in Fig. 6.

C. FRAMEWORK DETAILS

The research leverages the FL technique to develop a secure framework, using the proposed Edge-FL-based IDS to detect and classify various DDoS cyber threats. Fig. 7 shows the framework of the proposed edge-FL-based IDS. As shown in Fig. 7, each client trains the classifier with heterogeneous data and sends updated parameters to the central server. Consequently, the proposed Edge-FL-based IDS enables each client to improve security and privacy against DDoS threats.

D. IMPLEMENTATION DETAILS

1) ASSUMPTIONS

The following are a few notions that are employed in the research while building the proposed Edge-FL-based IDS:

- **Participant client (K):** The participant clients are selected randomly or based on availability. All clients are reliable and participate in the training process. The research considers $K = 3$, $K = 5$, and $K = 7$ participant clients that hold their datasets.
- **Dataset partition:** FL supports three types of data partitioning: horizontal, vertical, and hybrid [52], [53], [54]. The research utilized the 'Horizontal FL' approach, in which each participant client's dataset comprises

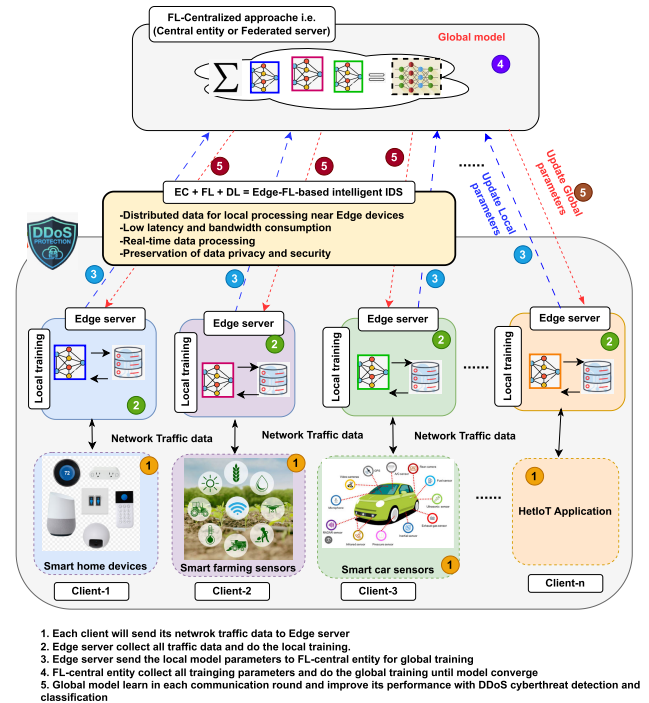


FIGURE 7. Framework for the proposed Edge-FL-based IDS.

different samples but the same independent feature space. For example, client i and client j contain different rows of data with the same features, and each row indicates the particular data for one particular target label, i.e., DDoS attack class. It is defined as [56] follows: Let the local dataset LD be presented in terms of the matrix to represent the data held by each participant client i . The matrix rows represent the number of samples S , and the columns represent the features X and label Y corresponding to the samples, then:

$$X_i = X_j, Y_i = Y_j, S_i \neq S_j, \forall LD_i, LD_j, i \neq j \quad (1)$$

- **Data distribution:** The data distribution in FL can be either IID or non-IID. The research addresses IID and non-IID data distribution using unbalanced datasets. In IID data distribution, each client holds all classes, whereas in non-IID data distribution, each client holds a different set of classes. The IID and non-IID data distribution details among various participant clients are shown in Fig. 10 and Fig. 17.
- **Global parameter:** The research adopted a homogeneous global model selection: All participant clients run the same classifier, a DL-based 1D-CNN classifier. The research uses the DL-based 1D-CNN classifier as a base model. Before the local training process starts, each participant client will get initial weights from this classifier, and then the local training will start using their local dataset.
- **Aggregator server:** The research used a centralized FL network structure in which a central entity is an

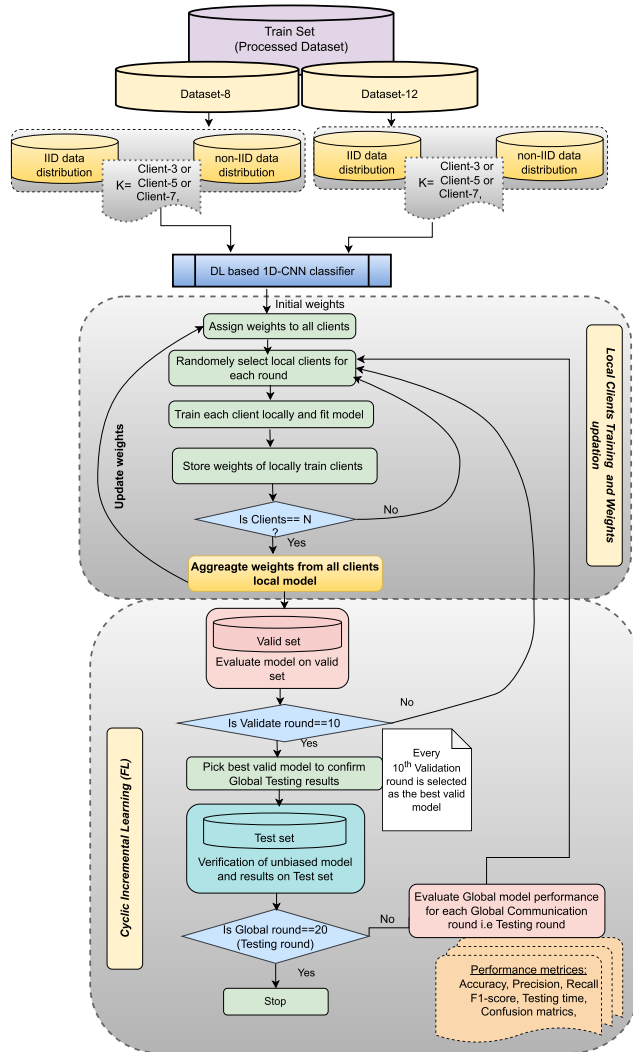


FIGURE 8. Flow diagram for the proposed Edge-FL-based IDS.

aggregator and uses the FedAvg algorithm to aggregate the global parameters [52], [54], [55]. It aggregates the trainable parameters sent by each participant client and sends newly updated parameters back to each participant client in an iterative manner until global model convergence occurs. An aggregator is responsible for sharing cyber threat knowledge globally without compromising legislation and keeping each participant's client updated regarding the new DDoS cyber threat trends.

The network flow diagram with implementation details of the proposed Edge-FL-based IDS is outlined in Fig. 8 and stated as follows:

- The research makes use of processed datasets: *Dataset-8* and *Dataset-12* to generate IID and non-IID data distributions, which are then randomly distributed among the participant clients: $K = 3$, $K = 5$, and $K = 7$.

Algorithm 2 proposed Edge-FL-Based IDS.

$r1$ = Global Communication Round (i.e. Testing round),
 $r2$ = Validation Round,
 K = Number of Participant Clients,
 C = Fraction of Clients Perform Training During Each Communication Round, i.e., $r1$ and $r2$,
 S = Size of the Data Samples With Input-Output Pair (X_i, Y_i) ,
 n_{r1} = Weight at Round $r1$
 R_{r1} = Set of Clients at Round $r1$

```

1 Central entity executes:
2 initial weights  $n$ 
3 for each  $r1 = 1$  to 20 do
4   for each  $r2 = 1$  to 10 do
5      $d \leftarrow \max(C.K, 1)$ 
6      $R \leftarrow$  random set of  $d$  clients
7     for each client  $k \in R$  in parallel do
8        $n_{r2+1}^k \leftarrow$  client-update ( $k, n_{r2}$ )
9     end
10     $n_{r2+1} \leftarrow \sum_{K=1}^K \frac{S_K}{S} n_{r2+1}^k$ 
11  end
12  Select the best validation model and do testing for
    $r1$  until the model converges
13 end
14
15 client-update ( $k, n_{r2}$ ):
16  $\beta \leftarrow$  Split  $LD_k$  into batches of  $B$ 
17 for each local-epoch  $i = 1$  to  $E$  do
18   for  $b \in \beta$  do
19      $n \leftarrow n - \text{ADAM}(\eta) \nabla l(n, b)$ 
20   end
21 end
22 end
23 return  $n$  to central entity

```

- The central entity, using the DL-based 1D-CNN classifier, initializes weights to all the participant clients, i.e., $K=3$, $K=5$, and $K=7$, to perform local training on their IID and non-IID data.
- All participant clients (K) will participate in the training process exactly once; however, which client initiates the training process first is determined randomly for each round. For example, if $K = 3$, then *client - 1*, *client - 2*, and *client - 3* will participate. If $K = 5$, then *client - 1*, *client - 2*, *client - 3*, *client - 4*, and *client - 5* will participate, and so on. Each participant client trains and fits the local model with its dataset. After fitting the local model, newly updated weights are sent to the central entity for aggregation.

- The research included *10-validation rounds* and *20-global communication rounds i.e. testing rounds*. Every 10th validation round is the improved model of the previous round. Hence, after every *10-validation rounds*, the research will select it as the best validation model [49], and its updated weights will be used for the testing, i.e., global communication round. The steps are repeated until the 20th global communication round.
- The *validation round* includes a separate validation set, and the *global communication round* includes a separate testing set for DDoS cyberthreats. The validation round aims to train each participant client *K* efficiently and ensure that the proposed Edge-FL-based IDS is not biased concerning any particular IID or non-IID dataset or any individual participant client. Each communication round helps the proposed Edge-FL-based IDS to enhance performance and achieve optimal accuracy. The proposed Edge-FL-based IDS pseudocode is presented in Algorithm 2.

IV. RESULTS AND DISCUSSIONS

The section evaluates the performance of the proposed Edge-FL-based IDS and Centralized IDS using 8 and 12-class classification. The section also compares the proposed Edge-FL-based IDS's performance with the existing state-of-the-art models. Further down, the simulation environment and performance indicators employed are discussed.

- **Simulation environment**
The simulation environment with hardware and software specifications are provided in Table 5.

TABLE 5. Simulation environment.

Software specification:	
Software tool	Spyder 5.0.1 IDE
Programming language	Python 3.7.16, Sklearn library, Matplotlib 3.5
API tool	TensorFlow 2.3, TensorFlow Federated 0.18
Hardware specification:	
Processor i7-10750H CPU @ 2.60GHz, 16GB RAM, Windows 10 64-bit OS, dedicated 4GB NVIDIA GEFORCE GTX 1650Ti	

- **Performance indicators**
The most widespread metrics used in the performance analysis of network intrusion detection are as follows [5], [18], [21]:

- Accuracy is used to calculate the percentage of correct classifications to the total number of instances, which is given by:

$$\frac{TP + TN}{TP + FN + TN + FP} \quad (2)$$

where *TP* computes the number of attack instances successfully classified as DDoS attacks. *FP* computes the number of benign instances falsely classified as DDoS attacks.

FN computes the number of attack instances falsely classified as benign.

TN computes the number of benign instances correctly classified as benign.

- Precision represents the ratio of correct intrusion classes to total expected intrusion results, which can be calculated as follows:

$$\frac{TP}{TP + FP} \quad (3)$$

- Recall represents the ratio of proper intrusion classes to the total number of samples that should have been identified as intrusions provided by,

$$\frac{TP}{TP + FN} \quad (4)$$

- F1-score calculates the harmonic mean of Precision and recall,

$$2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

- Confusion metrics is a metric that summarizes the performance of AI models on a set of test data. The number of correct and incorrect forecasts is summed up with count values and divided by class.

- **Performance analysis**

The research analyses the performance under two scenarios: Centralized IDS and the proposed Edge-FL-based IDS. The DL-based 1D-CNN classifier is a base classifier used for both scenarios. The performance is analyzed using Dataset-8 and Dataset-12. The hyperparameter settings used for both scenarios are provided in Table 6.

TABLE 6. Hyper-parameter settings for Centralized IDS and Edge-FL based intelligent IDS.

Hyperparameter Settings		Value
Base classifier's settings	Learning rate	0.01
	Optimizer	ADAM
	Activation function	Relu
	Classification function	SoftMax
	Batch Size	128
Centralized IDS	Loss function	Sparse_categorical_crossentropy
	Epoch	20
Edge-FL based intelligent IDS	Local epoch	1
	validation round	10
Global communication round (i.e. Testing rounds)		20

Scenario 1: In Centralized IDS, all traffic data is located at a central server and trained using a DL-based 1D-CNN classifier. The research considers the 20% validation set to ensure that the trained IDS is not biased and performs effectively by classifying various DDoS cyber threats. The implementation pseudocode is presented in Algorithm 3.

Scenario 2: In the proposed Edge-FL-based IDS, data is located at different clients, and each client is responsible for the local training of the data using a DL-based 1D-CNN classifier. The research examines three sets of client distributions, $K = 3$, $K = 5$, and $K = 7$, with

Algorithm 3 Centralized IDS.

For Dataset D , the Initial Weight Assigned Is n , η Is Learning Rate

```

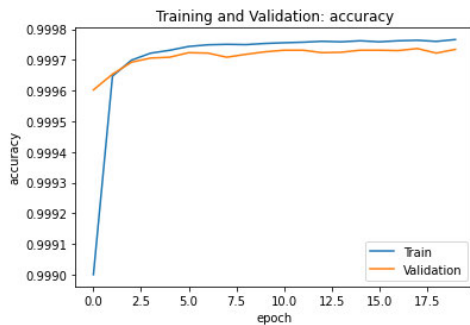
1 central entity (n):
2  $\beta \leftarrow$  Split  $D$  into batches of  $B$ 
3 for each epoch  $i=1$  to  $E$  do
4   for  $b \in \beta$  do
5      $n \leftarrow n - \text{ADAM}(\eta) \nabla l(n, b)$ 
6   end
7 end
8 end
9 return  $n$  to central entity

```

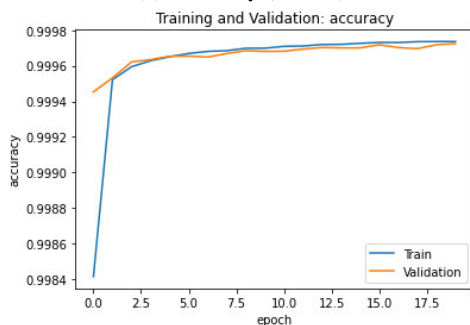
two data distributions, IID and non-IID, over 20-global communication rounds, i.e., testing round. Also, The research considers 10-validation rounds to validate the performance and select the best validation model to test the performance. It helps the research to enhance the training process and perform adequately.

A. CENTRALIZED IDS

Dataset-8 and Dataset-12 are employed for the assessment of the Centralized IDS performance. The evaluation results of the Centralized IDS for 8 class and 12 class classifications are provided in Fig. 9. The Centralized IDS achieves an overall accuracy of 99.97% for 8 class and 12 class classifications.



(a) Accuracy (8 class)



(b) Accuracy (12 class)

FIGURE 9. Performance of Centralized IDS.**B. PROPOSED EDGE-FL-BASED IDS**

The research analyzes the performance of the proposed Edge-FL-based IDS using Dataset-8 and Dataset-12 with participant clients $K = 3$, $K = 5$, and $K = 7$ under IID and non-IID data distribution. The performance assessment is discussed in depth below.

1) 8 CLASS

The details of the IID and non-IID data distributions among clients, $K = 3$, $K = 5$, and $K = 7$, are provided in Fig. 10. The research used a random distribution of DDoS traffic data across the participant clients. According to Fig. 10a, in IID data distribution, all participant clients, $K = 3$, $K = 5$, and $K = 7$, hold seven DDoS traffic data and one benign traffic data but different samples.

Further, as per Fig. 10b non-IID data distribution, all participant clients, $K = 3$, $K = 5$, and $K = 7$, hold a random set of DDoS attack traffic and benign traffic.

E.g., $K = 3$, *client-1* holds any four DDoS attack traffic, i.e., LDAP, NetBIOS, MSSQL, Portmap, and benign traffic data;

client-2 holds Syn, UDP, UDPLag, MSSQL, and benign traffic data; and *client-3* holds UDP, UDPLag, LDAP, NetBIOS, and Benign traffic data.

$K = 5$, *client-1*, *client-2*, *client-3*, *client-4* hold any three DDoS traffic data and benign data, whereas *client-5* contains two DDoS traffic data and benign data.

$K = 7$, *client-1*, *client-2*, *client-3*, *client-4*, *client-5*, *client-6*, and *client-7* hold any two DDoS traffic data and benign data.

Further, Table 7 provides the worst, best, and global performance analysis for 8 class IID and non-IID data distributions. As the training begins, all clients train the classifier with local data and build a global model. Its performance is validated using the valid set and tested against the test set. The 1st validation round (referred to as *Worse-case*), the 10th validation round (referred to as *Best-case*), and the 1st global communication rounds (referred to as *Global-case*) performances are reported in the Table 7.

a: PERFORMANCE ASSESSMENT FOR 8 CLASS NON-IID

The performance analysis for 8 class is depicted in Table 7, and Figs. 11, 12, and 13 for non-IID data distribution.

As shown in Table 7 $K=3$, the 1st validation round accuracy is 46.91%, which continues to improve, and it reaches 98.87% on the 10th validation round. As previously stated, after the first ten validation rounds, the weights of the best validation rounds are used to test the performance of the global communication round. Fig. 11 illustrates the best validation round performance with accuracy and loss. As shown in Fig. 12, the 1st global communication round achieved 98.88% accuracy with a loss of 0.0294. Following that, during each global communication round, the proposed Edge-FL-based IDS learns appropriately and performs well with continual improvement with the support of every ten

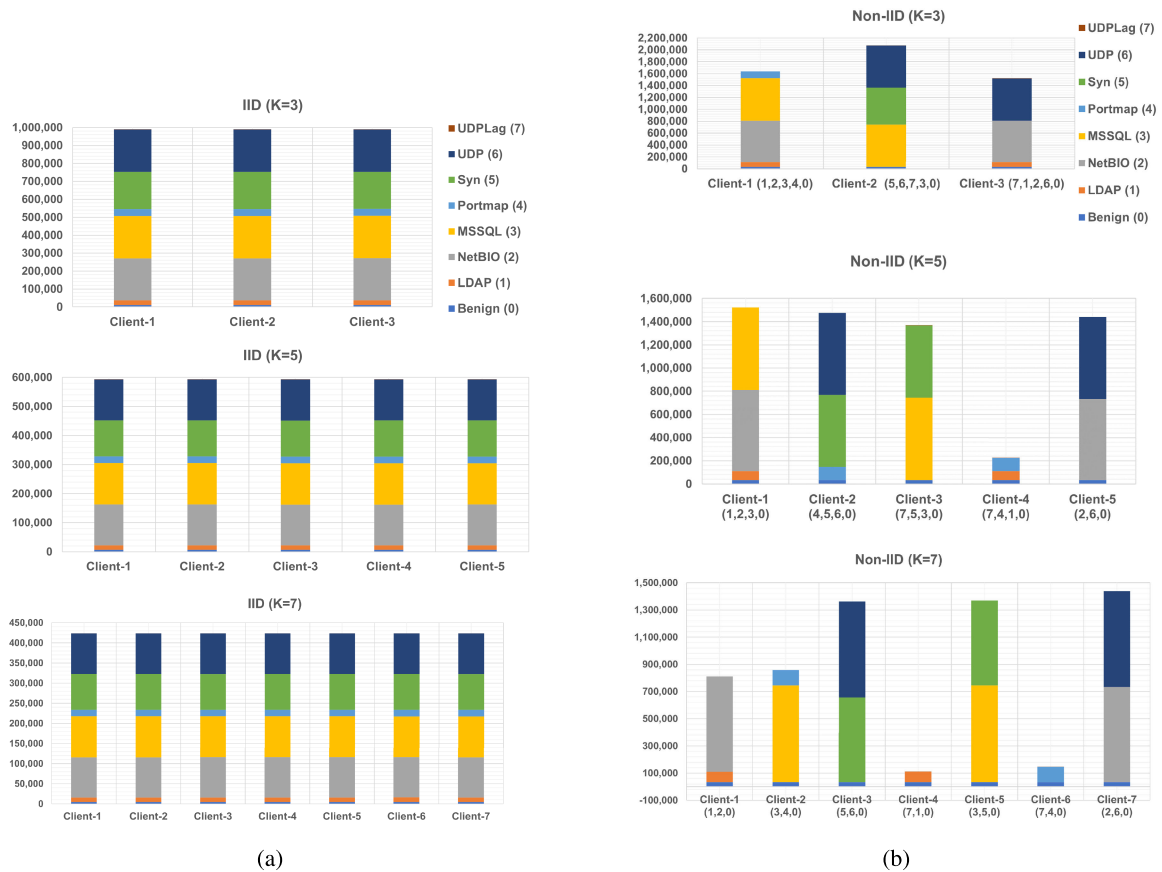


FIGURE 10. Data distribution for 8 class classification (a) IID and (b) NonIID.

TABLE 7. Performance analysis: Worse-case, Best-case and Global-case.

Participant clients	8 class						12 class					
	non-IID			IID			non-IID			IID		
	Worse (1 st round)	Best(10 th round)	Global(1 st round)	Worse (1 st round)	Best(10 th round)	Global(1 st round)	Worse (1 st round)	Best(10 th round)	Global(1 st round)	Worse (1 st round)	Best(10 th round)	Global(1 st round)
K=3	46.91%	98.87%	98.88%	99.71%	99.97%	99.97%	12.23%	62.78%	62.85%	99.77%	99.91%	99.95%
K=5	17.19%	97.15%	97.19%	99.69%	99.96%	99.96%	4.70%	46.67%	46.89%	99.61%	99.86%	99.95%
K=7	16.85%	70.31%	70.34%	99.57%	99.95%	99.94%	1.45%	21.80%	19.71%	99.69%	99.70%	99.93%

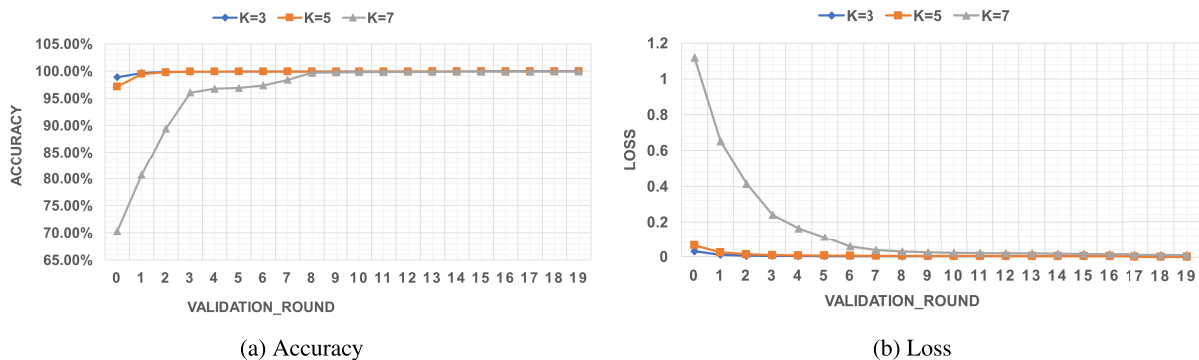


FIGURE 11. Best Validation round performance for non-IID (8 class).

validation rounds (refer to Fig. 11). Also, as illustrated in Fig. 12, after the 12th global communication round, the

proposed IDS had reached an accuracy of 99.97% and started converging.

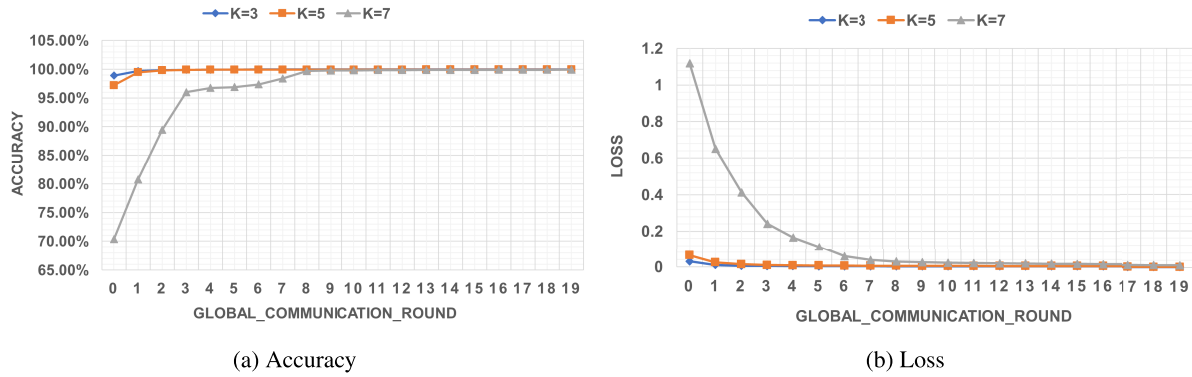


FIGURE 12. Global Communication round for non-IID (8 class).

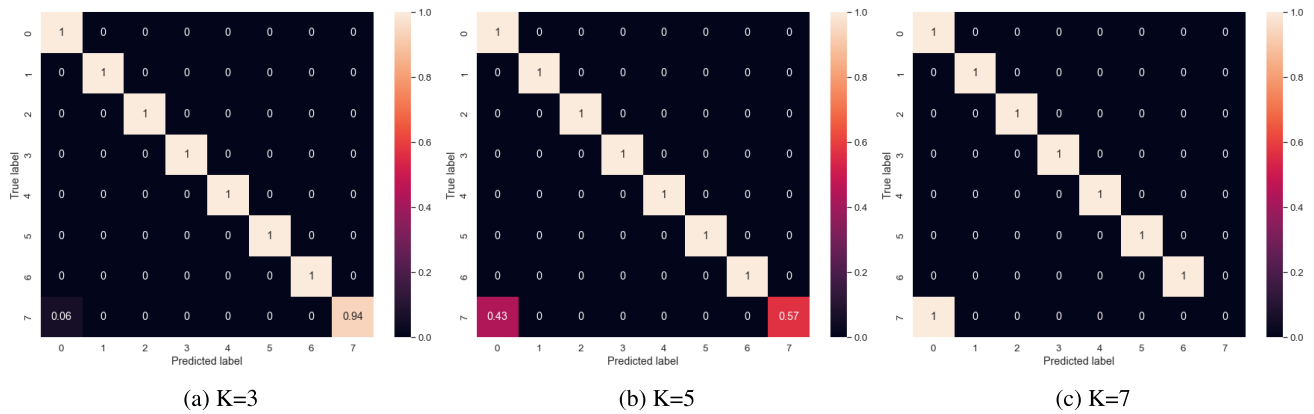


FIGURE 13. Confusion metrics for non-IID (8 class).

Similarly, for $K = 5$, as shown in Table 7, during the 1st validation round, achieve 17.19% accuracy and reaches 97.15% at the 10th round. The 1st global communication round attained the same accuracy as the 10th validation round, i.e., 97.17%, the best validation round. Then, the classifier keeps on learning during each global communication round while preserving data with individual clients and reaches 99.95% at the 14th global communication round; after that, its performance is uniform, as highlighted in Fig. 12.

For $K=7$, the classifier trains the model with seven clients and reaches the lowest accuracy, 16.85%, during the 1st validation round. Each validation round advances performance, and the 10th validation round obtains 70.31% accuracy. The 1st global communication round performs similarly to the best case. Fig. 12a shows the global round performance continues to improve with minor tweaks during rounds 4th, 5th, 6th, 7th, 8th. Lastly, following the 10th global communication round, performance remains consistent with an accuracy of 99.01%.

The losses during the best validation and global communication rounds are shown in Fig. 11b and Fig. 12b, respectively. The confusion metrics for $K=3$ (Fig. 13a), $K=5$ (Fig. 13b), and $K=7$ (Fig. 13c), are shown in Fig. 13. Research demonstrates that the proposed Edge-FL-based IDS is not biased and performs admirably

on non-IID data distribution with *cyclic incremental learning*.

b: PERFORMANCE ASSESSMENT FOR 8 CLASS IID

The performance evaluation for 8 class IID data distribution is shown in Table 7, Figs. 14, 15, and 16.

As shown in Table 7, $K=3$, 1st validation round achieves a 99.71% outcome that keeps continuing, and at the 10th round, it reaches 99.97%. The aggregated trainable parameters are validated further to select the best validation model, and its performance is presented in Fig. 14. The 1st global communication round attained a 99.97% outcome, as shown in Fig. 15. The continuous learning process of the proposed IDS aids in accurately detecting and classifying the DDoS cyber threat. The confusion metrics are shown in Fig. 16a.

Similarly, for $K=5$, the proposed Edge-FL-based IDS displays consistent performance and convergence after 12th global communication rounds, with 99.97% as the best accuracy. The confusion matrices in Fig. 16b show that the proposed IDS successfully classifies all DDoS classes.

$K=7$ clients need to perform more adequately compared to $K=3$ and $K=5$. It achieves 99.94% accuracy in the 1st global communication round and up to 99.96% accuracy in subsequent rounds. The losses experienced during the best validation and global communication rounds are shown in

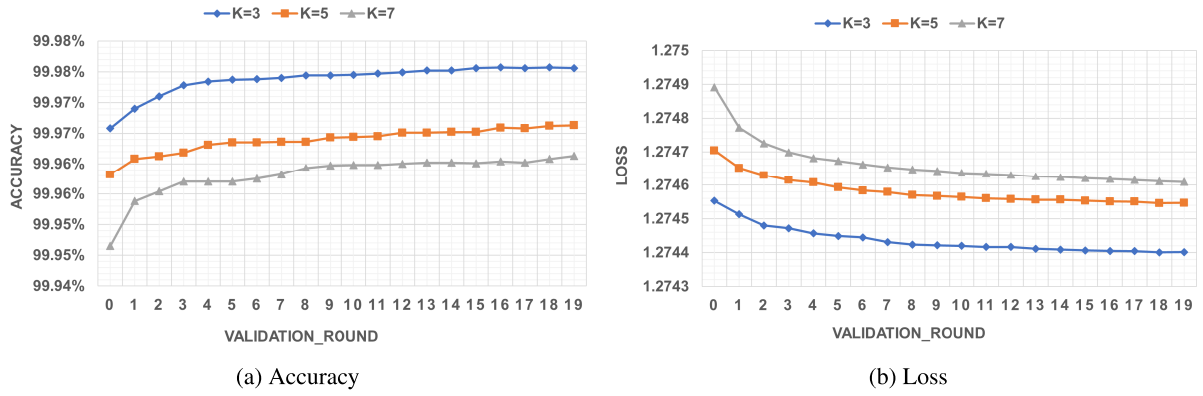


FIGURE 14. Best Validation round performance for IID (8 class).

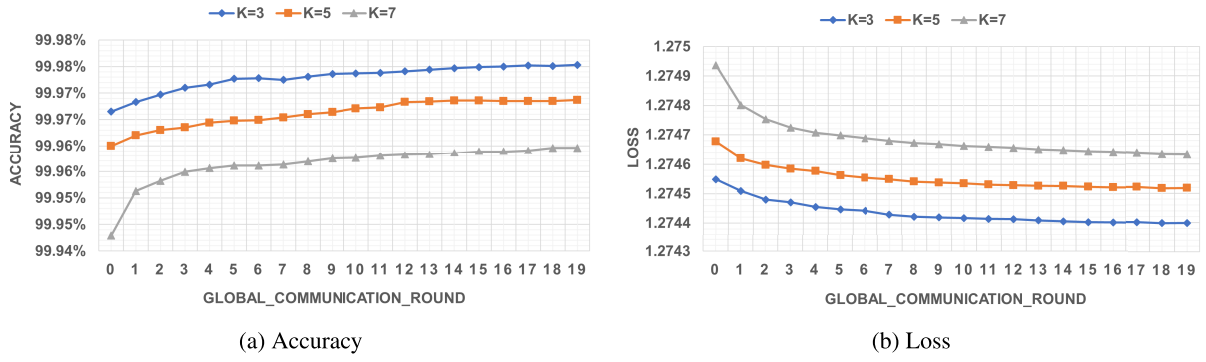


FIGURE 15. Global Communication round performance for IID (8 class).

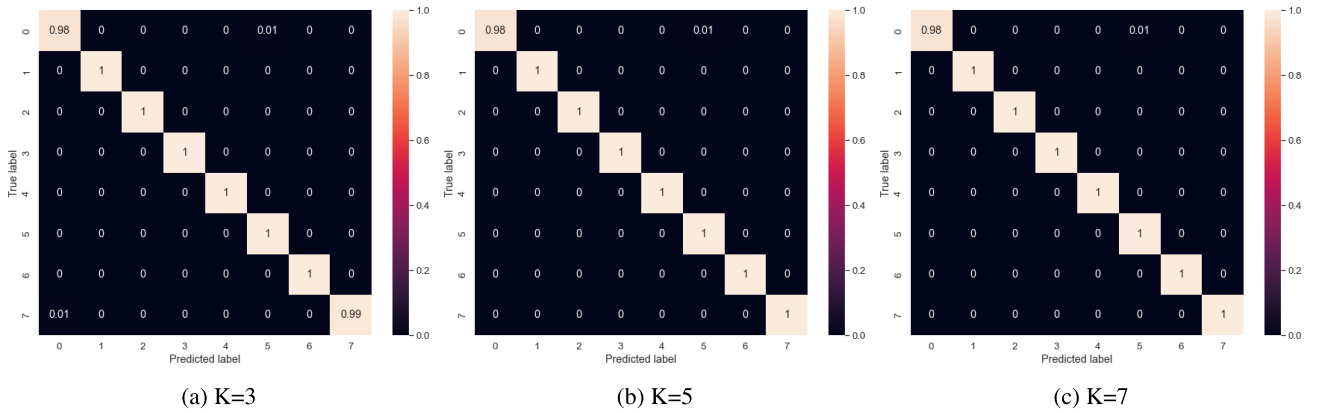


FIGURE 16. Confusion metrics for IID (8 class).

Fig. 14b and Fig. 15b, respectively. Furthermore, as demonstrated in Fig. 15, $K=7$ sustains a significant loss compared to $K=3$ and $K=5$, i.e., 1.2747, 1.2744, and 1.2746, respectively. The confusion metrics in Fig. 16c reveal that the classifier misidentifies a few benign traffic samples as UDPLag traffic.

2) 12 CLASS

The details of the IID and non-IID data distributions among the participant clients, $K=3$, $K=5$, and $K=7$, are provided in Fig. 17.

According to Fig. 17a, in IID data distribution, each participant client, $K=3$, $K=5$, and $K=7$, holds eleven DDoS traffic data and one benign traffic data. Further, Fig. 17b non-IID data distribution, each participant client, $K=3$, $K=5$, and $K=7$, holds a random set of attack and benign traffic.

E.g., $K=3$, each *client-1*, *client-2*, and *client-3* holds five DDoS and one benign traffic data.

$K=5$, each client holds random three attack data and benign traffic, i.e.,

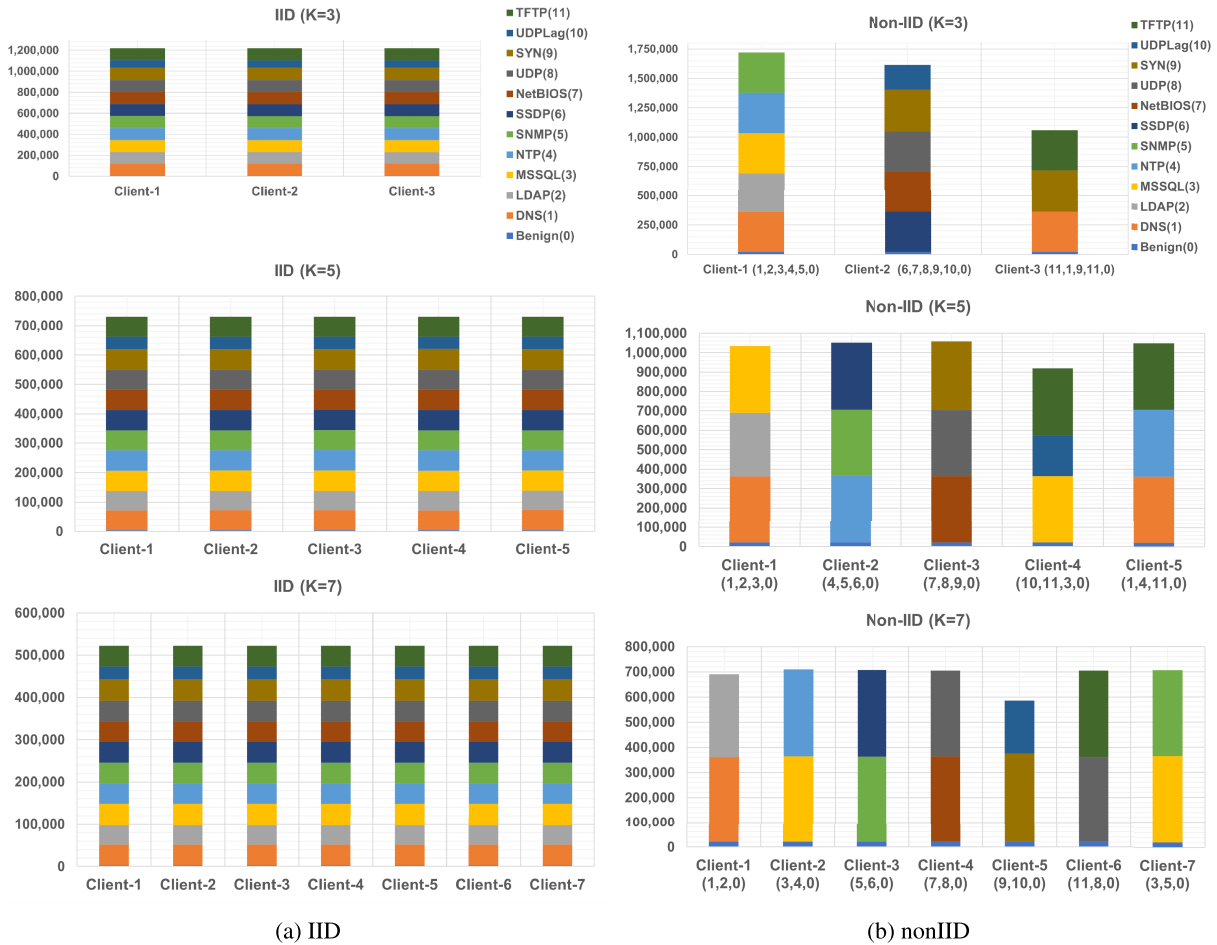


FIGURE 17. Data distribution for 12 class classifications.

client-1: LDAP, DNS, MSSQL, and benign data;
client-2: NTP, SNMP, SSDP, and benign data;
client-3: NetBIOS, UDP, Syn and benign data;
client-4: UDPLag, TFTP, MSSQL, and benign data;
client-5: DNS, NTP, TFTP, and benign data.
 $K = 7$, each client holds two DDoS attacks and benign data.

α : PERFORMANCE ASSESSMENT FOR 12 CLASS NON-IID

The performance analysis for 12 class classification is depicted in Table 7, Figs. 18, 19, and 20.

For $K=3$, according to Table 7, the training begins with a worse accuracy rate of 12.23%. It improves with each validation round, and it reaches 62.78% after the 10th validation round. The best validation round's updated weights are used to evaluate the global communication round. Fig. 18 illustrates the best validation round performance with accuracy and loss. The 1st global communication round achieved 62.85% accuracy with a 2.204 loss. After the 14th global communication round, the classifier began converging, and the global accuracy achieved is 99.07%.

Next, $K=5$, the 1st validation round accuracy reported is 4.70% (refer to Table 7.), which improves after every

validation round. Fig. 18 shows the best validation accuracy achieved by the proposed Edge-FL-based IDS during each 10th round. Each participant's client holds different DDoS traffic data. However, the classifier learns from the trainable parameters during each round and finally reaches a 96.44% accuracy rate at the 20th global communication round. Fig. 19 presents the performance assessment during each global communication round.

Further $K=7$, as shown in Table 7, the 1st validation round for seven different participant clients with heterogeneous data distribution achieves 1.45% accuracy. However, the EDA and hyperparameter settings help the proposed Edge-FL-based IDS perform well by increasing the detection accuracy rate. As indicated in Fig. 19, the best-reported accuracy is 52.39% with continuous loss reduction. However, it performs poorly among the other participant clients, $K = 3$ and $K = 5$.

The losses during the best validation and global communication rounds are shown in Fig. 18b and Fig. 19b, respectively. The confusion matrix for $K = 3$, $K = 5$, and $K = 7$ is provided in Fig. 20a, Fig. 20b, Fig. 20c respectively. The confusion matrix (Fig. 20c) for $K = 7$ indicates that the classifier misinterprets DDoS traffic as benign, impacting

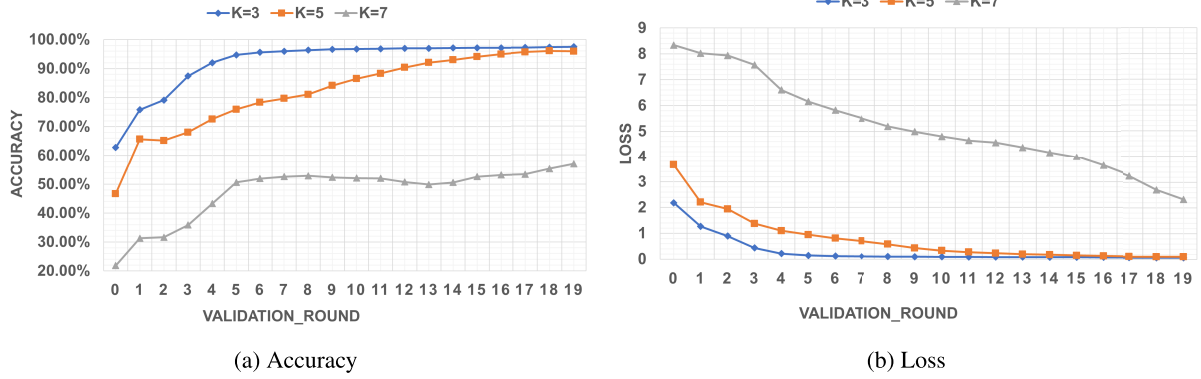


FIGURE 18. Best Validation round performance for non-IID (12 class).

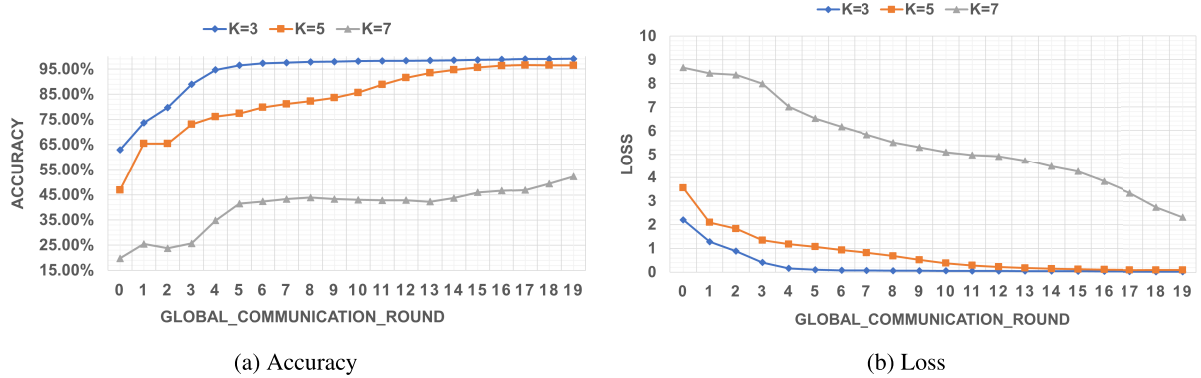


FIGURE 19. Global Communication round performance for non-IID (12 class).

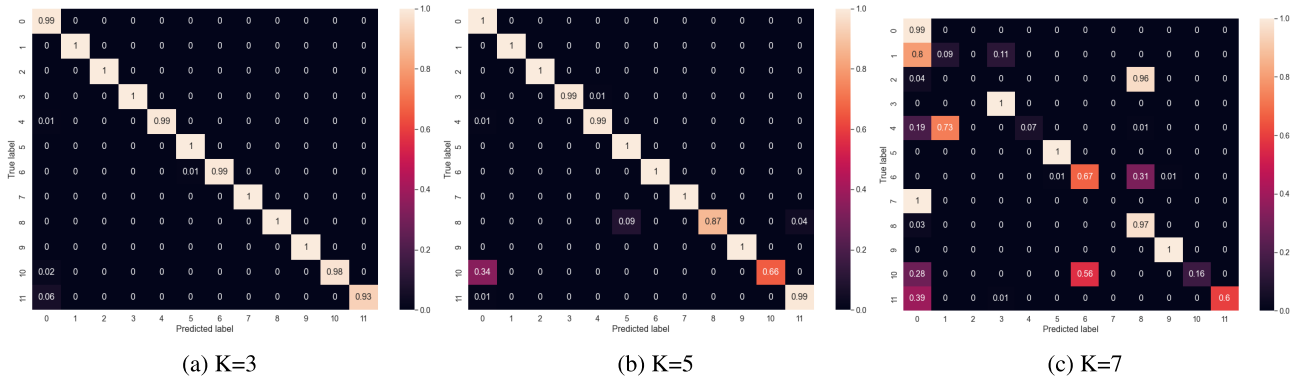


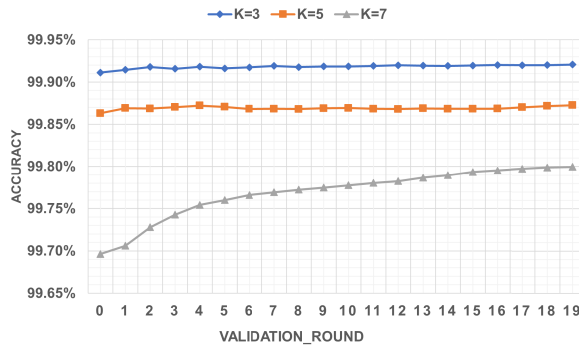
FIGURE 20. Confusion metrics for non-IID (12 class).

overall performance. The reason is each client holds a diverse set of traffic data. Fig. 18 and Fig. 19 show that the proposed IDS continues to learn with each round with modest accuracy fluctuation. Further hyperparameter tuning and more communication rounds will improve its performance. In contrast, K=3 and K=5 began converging after the 16th round. The research findings indicate that the proposed Edge-FL-based IDS performs satisfactorily with K=3 and K=5, excluding K=7 with non-IID distribution.

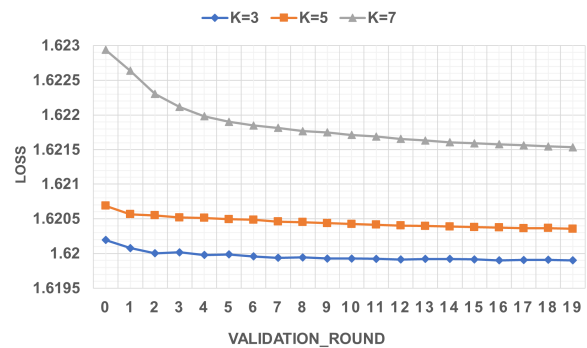
b: PERFORMANCE ASSESSMENT FOR 12 CLASS IID

The 12 class IID data distribution performance evaluation is shown in Table 7, Figs. 21, 22, and 23.

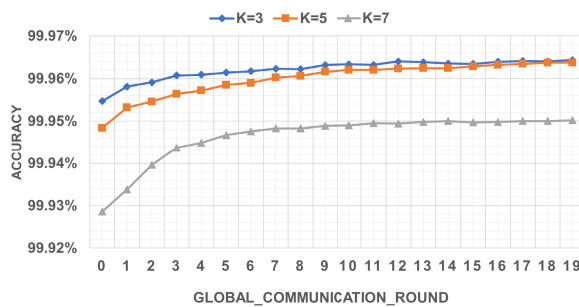
As illustrated in Table 7 K=3, the 1st validation round achieves 99.76% and continues until the 10th round achieves 99.91%. The best validation model performance with accuracy and loss is presented in Fig.21. The proposed IDS continuous learning process assisted in achieving 99.96% accuracy at the 4th global communication round



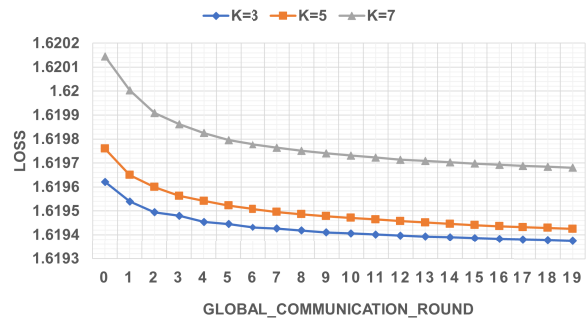
(a) Accuracy



(b) Loss

FIGURE 21. Best Validation round performance for IID (12 class).

(a) Accuracy



(b) Loss

FIGURE 22. Global Communication round performance for IID (12 class).

(refer to Fig. 22), following which the proposed Edge-FL-based IDS began converging with minor accuracy variations.

Similarly, for $K = 5$, as shown in Fig. 21 during 1st validation round, the classifier achieves 99.86% as the best validation accuracy and loss as 1.620. The proposed Edge-FL-based IDS displays consistent performance and convergence after 3rd global communication rounds, with 99.96% as the global accuracy (refer to Fig. 22). The confusion metrics in Fig. 23b show that the proposed IDS successfully classifies all DDoS classes with a 100% detection rate.

As indicated in Fig. 22 $K = 7$ achieves 99.92% accuracy in the 1st global communication round and up to 99.95% accuracy in subsequent rounds, i.e., 5th. After 5th, the global communication round classifier demonstrates consistent gain. The confusion metrics in Fig. 23a, Fig. 23b, Fig. 23c also represent that each participant client $K = 3$, $K = 5$, and $K = 7$ successfully detects and classifies various DDoS and benign traffic data. Although the samples at $K = 3$, $K = 5$, and $K = 7$ are different, the proposed Edge-FL-based IDS learns and settles quickly compared to the non-IID data distribution. The reason is that even though samples are different, each client is familiar with every DDoS class trend, which helps the proposed IDS to perform better.

TABLE 8. Comparative performance evaluation with state-of-the-art IDS.

Year	AI classifier & Ref.	Communication round	Accuracy (%)
2021	GRU [38]	2000, 10	98
2021	CNN [35]	600	99.2
2021	GBDT [33]	—	65.04, 64.52
2022	MLP [31]	605	98.53
2022	MLP [32]	30	98.84
2022	CNN-MLP [37]	—	98.18, 98.37

C. OVERALL PERFORMANCE DISCUSSION AND OBSERVATION

The overall performance of the Centralized IDS and the proposed Edge-FL-based IDS is presented in Table 9. The performance analysis of the proposed Edge-FL-based IDS with existing state-of-the-art IDS is depicted in Table 8. The existing state-of-the-art IDS [31], [32], [33], [35], [37], [38] uses the same CICDDoS2019 security dataset with different AI classifiers. The research observations concerning performance evaluation with Centralized IDS and existing state-of-the-art IDS are discussed as:

Observation-1: The Centralized IDS achieves 99.97% accuracy for 8-class and 12-class classification. The Centralized IDS and the proposed Edge-FL-based IDS performance

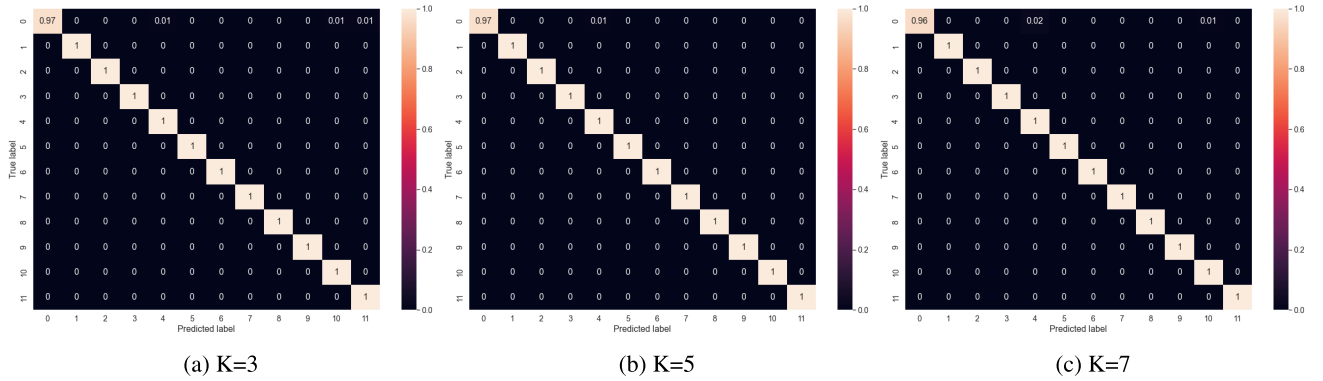


FIGURE 23. Confusion metrics for IID (12 class).

TABLE 9. Overall comparative performance (Testing accuracy).

	Centralized IDS		Edge-FL based intelligent IDS											
Performance evaluation			8 class						12 class					
	12 class	8 class	IID			non-IID			IID			non-IID		
			Client-3	Client-5	Client-7	Client-3	Client-5	Client-7	Client-3	Client-5	Client-7	Client-3	Client-5	Client-7
Accuracy (%)	99.97	99.97	99.98	99.97	99.96	99.97	99.94	99.90	99.96	99.96	99.95	99.07	96.44	52.39

are comparable. However, concerning Centralized IDS, every HetIoT device must exchange traffic data with a central entity or third party, which raises privacy concerns and violates GDPR legislation guidelines [21], ultimately causing security issues. On the other hand, the proposed Edge-FL-based IDS addresses this concern by only sharing trainable parameters with a central entity. It further ensures security & privacy by processing data and model near the edge network.

Observation-2: The proposed Edge-FL based IDS outperforms the existing state-of-the-art IDS [31], [32], [33], [35], [37], [38]. As shown in Table 8, state-of-the-art IDS using MLP [32] considers a balanced dataset with IID distribution and four participant client. In comparison, the proposed Edge-FL-based IDS considers an unbalanced dataset with IID and non-IID distribution and diverse set of participant client. The unbalanced dataset with heterogeneous data distribution is challenging. The research shows that the proposed Edge-FL-based IDS performs better when addressing the concern.

For 12 class classifications, the participant clients $K=3$, $K=5$, and $K=7$ achieved 99.96%, 99.96%, and 99.95% accuracy rates with IID data distribution. Furthermore, using non-IID data distribution, the participant clients $K=3$, $K=5$, and $K=7$ achieved an accuracy of 99.07%, 96.44%, and 52.39%, respectively.

For 8 class classifications, the participant client $K=3$, $K=5$, and $K=7$ achieved 99.98%, 99.97%, and 99.96% accuracy rates with IID data distribution. Further, with non-IID data distribution, the participant client $K=3$, $K=5$, and $K=7$ achieved 99.97%, 99.94%, and 99.90% accuracy, respectively.

Observation-3: Despite the proposed Edge-FL-based IDS, for 12 class non-IID data distribution using $K=7$, performance is unsatisfactory; it excels with $K=3$ and $K=5$. Moreover, the 12-class IID distribution performance with $K=3$, $K=5$, and $K=7$ is satisfactory. Next, the 8-class non-IID and IID distribution performance is good. The proposed Edge-FL-based IDS performs better with unbalanced non-IID and IID distributions. The research adopted data preprocessing, DL-based 1D-CNN classifier, and hyperparameter settings helps the proposed Edge-FL-based IDS perform effectively. It successfully detects and classifies DDoS threats by preserving privacy concerns and boosting HetIoT security.

V. CONCLUSION

The research presented an Edge-FL-based intelligent IDS to defend HetIoT applications against DDoS cyber attacks. The proposed Edge-FL-based IDS successfully detects and classifies various DDoS attacks while maintaining security and data privacy. The research developed a DL-based 1D-CNN classifier is a simple, lightweight, and less complex. The research analyzes the performance of the Centralized IDS and the proposed Edge-FL-based IDS scenarios using Dataset-8 and Dataset-12. The proposed Edge-FL-based IDS detects and classifies DDoS cyber threats with the following accuracy:

(a) 8-class IID, $K=3$ is 99.98%, $K=5$ is 99.97%, $K=7$ is 99.96%. (b) 8-class non-IID, $K=3$ is 99.97%, $K=5$ is 99.94%, $K=7$ is 99.90%. (c) 12-class IID, $K=3$ is 99.96%, $K=5$ is 99.96%, $K=7$ is 99.95%. (d) 12-class non-IID, $K=3$ is 99.07%, $K=5$ is 96.44%, $K=7$ is 52.39%. Compared to Centralized IDS and the existing state-of-the-art

IDS, the proposed Edge-FL-based IDS performs well under IID and non-IID data distribution.

The research's limitation is that while the proposed Edge-FL-based IDS outperforms, its efficiency, in some cases, is unsatisfactory. However, increasing the number of communications rounds and fine-tuning the hyperparameters will improve its performance and be left as future work. Furthermore, the research can be expanded by employing the proposed hierarchical Edge-FL-based IDS, considering centralized and distributed on-site techniques.

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SHALAKA S. MAHADIK (Student Member, IEEE) received the B.E. and M.E. degrees in computer engineering from Mumbai University, India, in 2005 and 2011, respectively. She is currently pursuing the Ph.D. degree with the Birla Institute of Technology and Science, Pilani, Dubai Campus, Dubai, United Arab Emirates. She was a Lecturer, from 2005 to 2011, and an Assistant Professor, from 2011 to 2014. She has participated in and coordinated various workshops and national conferences. She published her research work in well-known, reputed conferences and journals, such as IEEE, Springer, and Elsevier. Her research interests include networks and security, the Internet of Things (IoT), edge computing, and artificial intelligence. Her research is currently based on security and privacy in heterogeneous IoT (HetIoT). During the Ph.D. degree, she won awards for the best paper presentation and the best research proposal.



PRANAV M. PAWAR (Member, IEEE) received the degree in computer engineering from Dr. Babasaheb Ambedkar Technological University, Maharashtra, India, in 2005, the master's degree in computer engineering from Pune University, in 2007, and the Ph.D. degree in wireless communication from Aalborg University, Denmark, in 2016. He was an Associate Professor with MIT ADT University, Pune, India, from 2018 to 2019, and an Associate Professor with the Department of Information Technology, STES's Smt. Kashibai Navale College of Engineering, Pune, from 2008 to 2018. From 2006 to 2007, he was a System Executive with POS-IPC, Pune. He is currently an Assistant Professor with the Department of Computer Science, Birla Institute of Technology and Science (BITS), Pilani, Dubai. Before BITS Pilani, he was a Postdoctoral Fellow with Bar-Ilan University, Israel, from March 2019 to October 2020, in the area of wireless communication and deep learning. He is an IBM DB2 and IBM RAD certified professional and completed NPTEL certification in different subjects. He received recognition from Infosys Technologies Ltd. for his contribution to the Campus Connect Program and also received different funding for research and attending conferences at the international level. He published more than 40 papers at the national and international levels. His research interests include security and privacy, machine and deep learning applications, reinforcement learning, the Internet of Things, and bioinformatics. He was a recipient of an Outstanding Postdoctoral Fellowship from Israel Planning and Budgeting Committee. His Ph.D. thesis received a nomination for the Best Thesis Award from Aalborg University.



RAJA MUTHALAGU was a Postdoctoral Research Fellow with the Air Traffic Management Research Institute, Nanyang Technological University, Singapore, from 2014 to 2015. He is currently an Associate Professor with the Birla Institute of Technology and Science, Pilani, Dubai Campus, Dubai, United Arab Emirates. His research interests include wireless communication, signal processing, aeronautical communication, and cyber security. He was a recipient of the Canadian Commonwealth Scholarship Award for the Graduate Student Exchange Program at the Department of Electrical and Computer Engineering, University of Saskatchewan, Saskatoon, SK, Canada, in 2010.

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